

EXP NO: 1

STUDY OF LOGIC GATES

DATE:

30/09/22

AIM:

To study about logic gates and verify the truth table.

APPARATUS REQUIRED:

SNO	COMPONENT	SPECIFICATION	QUANTITY
1	AND gate	IC 7408	1
2	OR gate	IC 7432	1
3	NOT gate	IC 7404	1
4	NAND gate 2 I/P	IC 7400	1
5	NOR gate	IC 7402	1
6	X-OR gate	IC 7486	1
7	NAND gate 3 I/P	IC 7410	1
8	IC Trainer kit	-	1
9	Path card	-	14

THEORY:

(*) Circuit that takes the logical decision and the process are called logic gates.

(*) Each gate has one or more input and only one output.

(*) OR, AND, and NOT are basic gates.

(*) NAND, NOR and EX-OR are known as universal gates.

(*) Basic gates from these gates.

PROCEDURE:

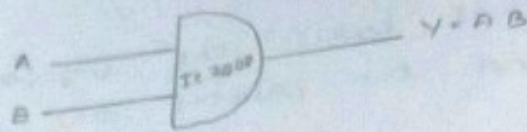
(*) Connection are given as per the circuit diagram.

(*) Logical input are given as per circuit diagram.

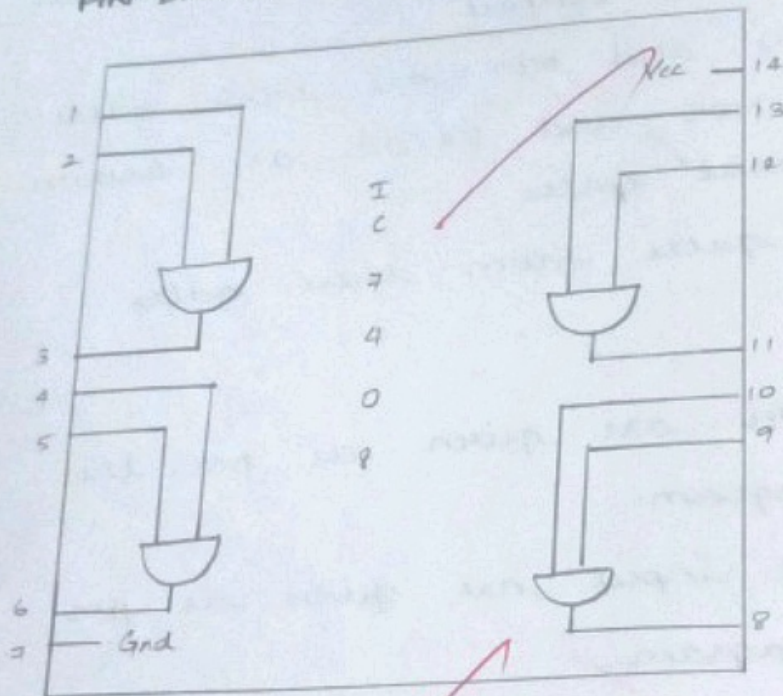
(*) Observe the output and verify the truth table.

AND GATE

SYMBOL:



PIN DIAGRAM:



TRUTH TABLE:

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

AND gate:

↳ The AND

multiplication

AND function

↳ The

the input

is low

the input

AND Gate:

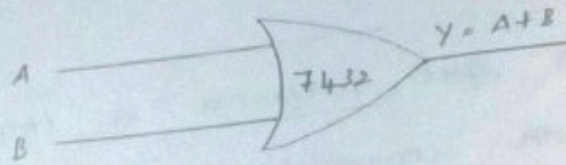
(*) The AND gate performs a logical multiplication commonly known as AND function.

(*) The output is high when both the inputs are high, the output is low level when only one of the inputs is low.

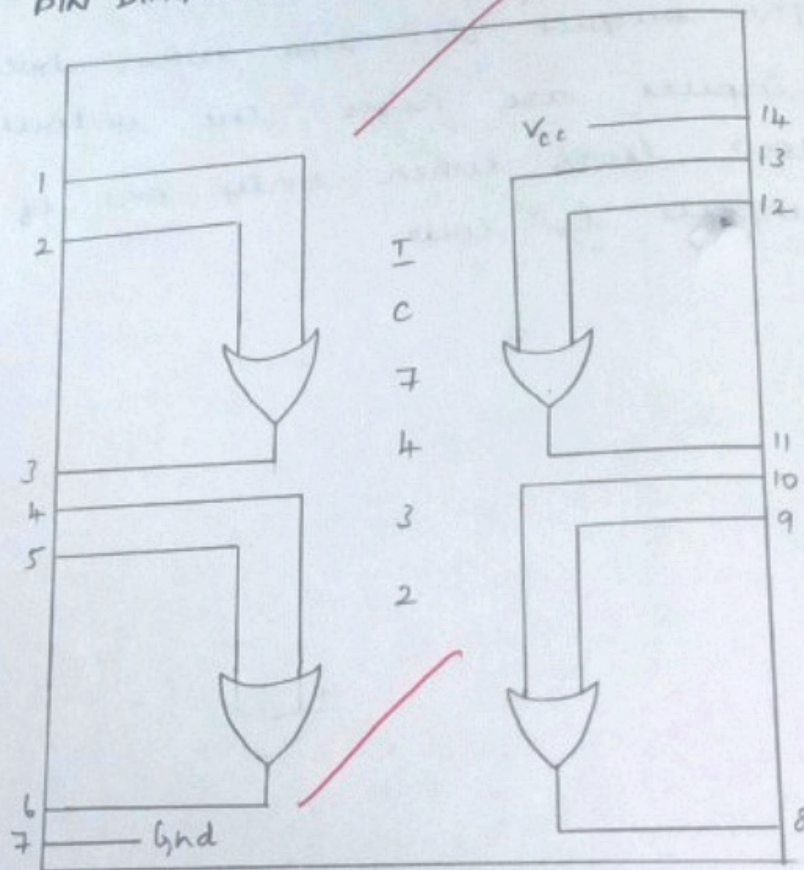


OR GATE

SYMBOL:



PIN DIAGRAM:



TRUTH TABLE:

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

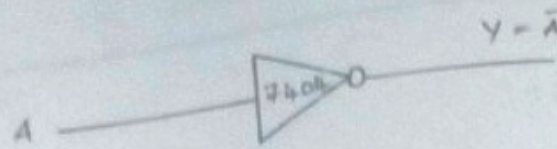
OR gate:

(*) The OR gate performs a logical addition commonly known as OR function.

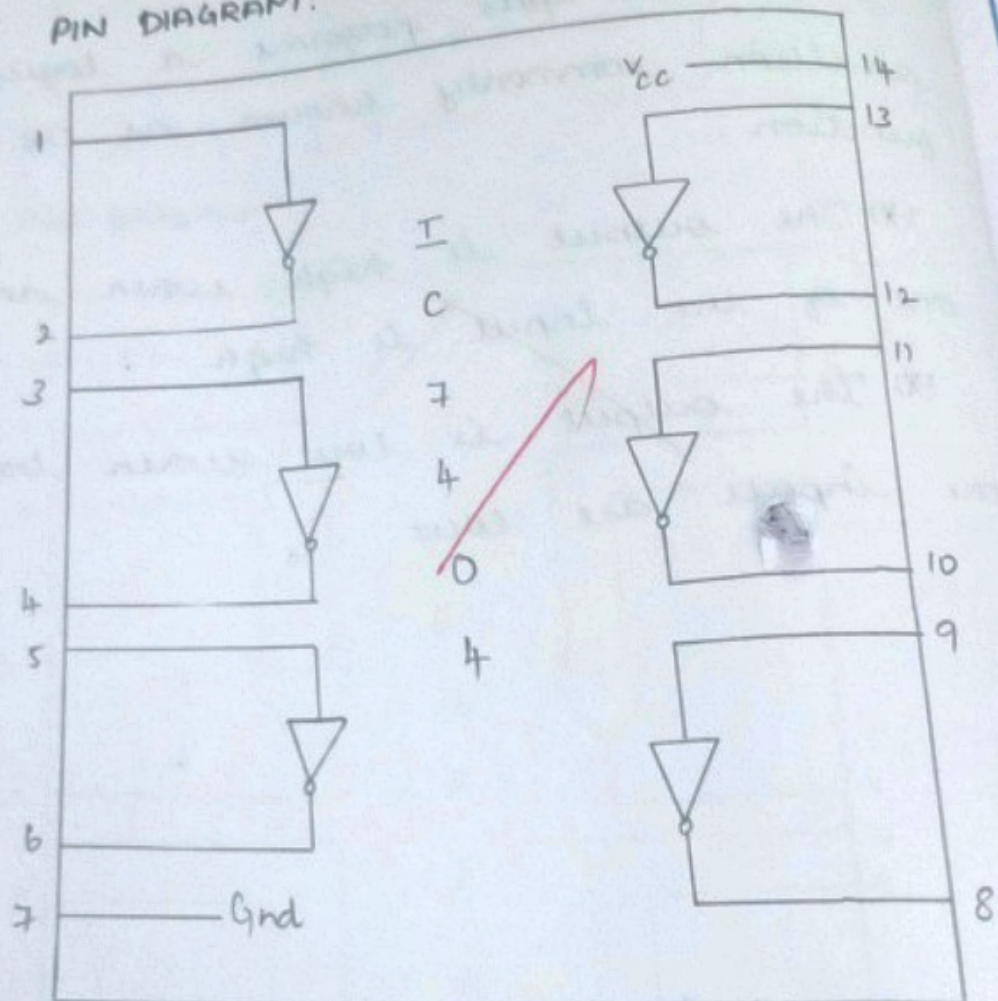
(*) The output is high when any one of the input is high.

(*) The output is low when both the inputs are low.

NOT GATE



PIN DIAGRAM:



TRUTH TABLE:

A	$\bar{A}$
0	1
1	0

NOT gate:

✶ The NOT gate is called an inverter.

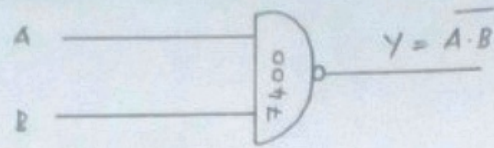
✶ The output is high when the input is low.

✶ The output is low when the input is high.

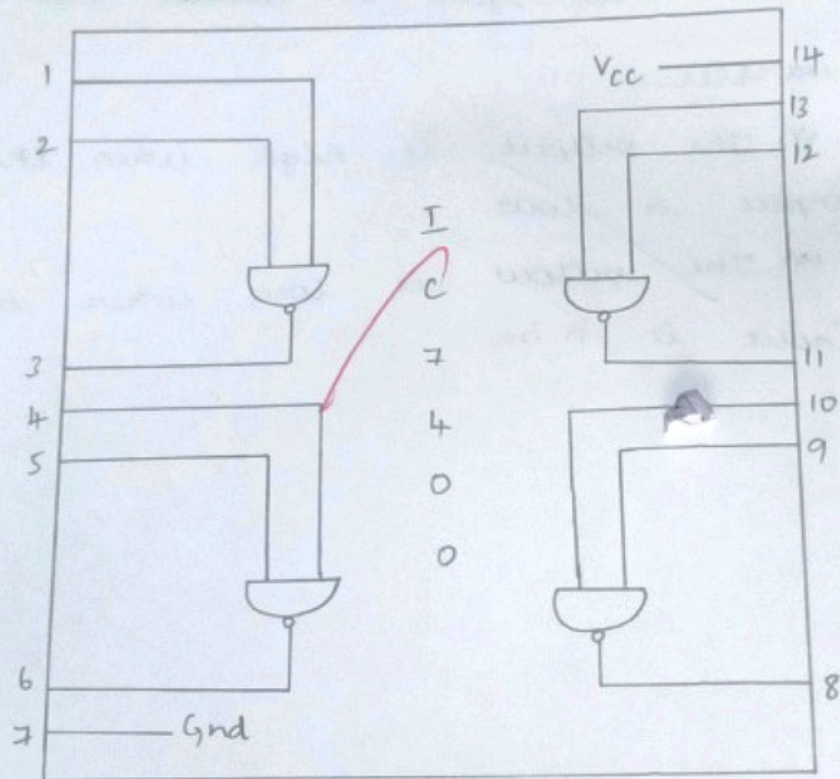
Input	Output
1	0
0	1

NAND GATE

SYMBOL:



PIN DIAGRAM:



TRUTH TABLE

A	B	$Y = A \cdot \overline{B}$
0	0	1
0	1	1
1	0	1
1	1	0

NAND gate:

(*) The NAND gate is a contraction of AND-NOT.

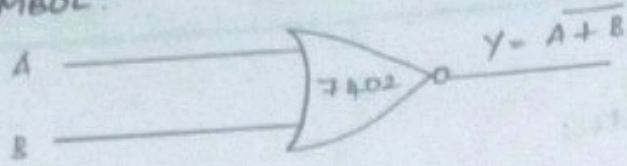
(*) The output is high when the both inputs are low.

(*) The output is low when both inputs are high.

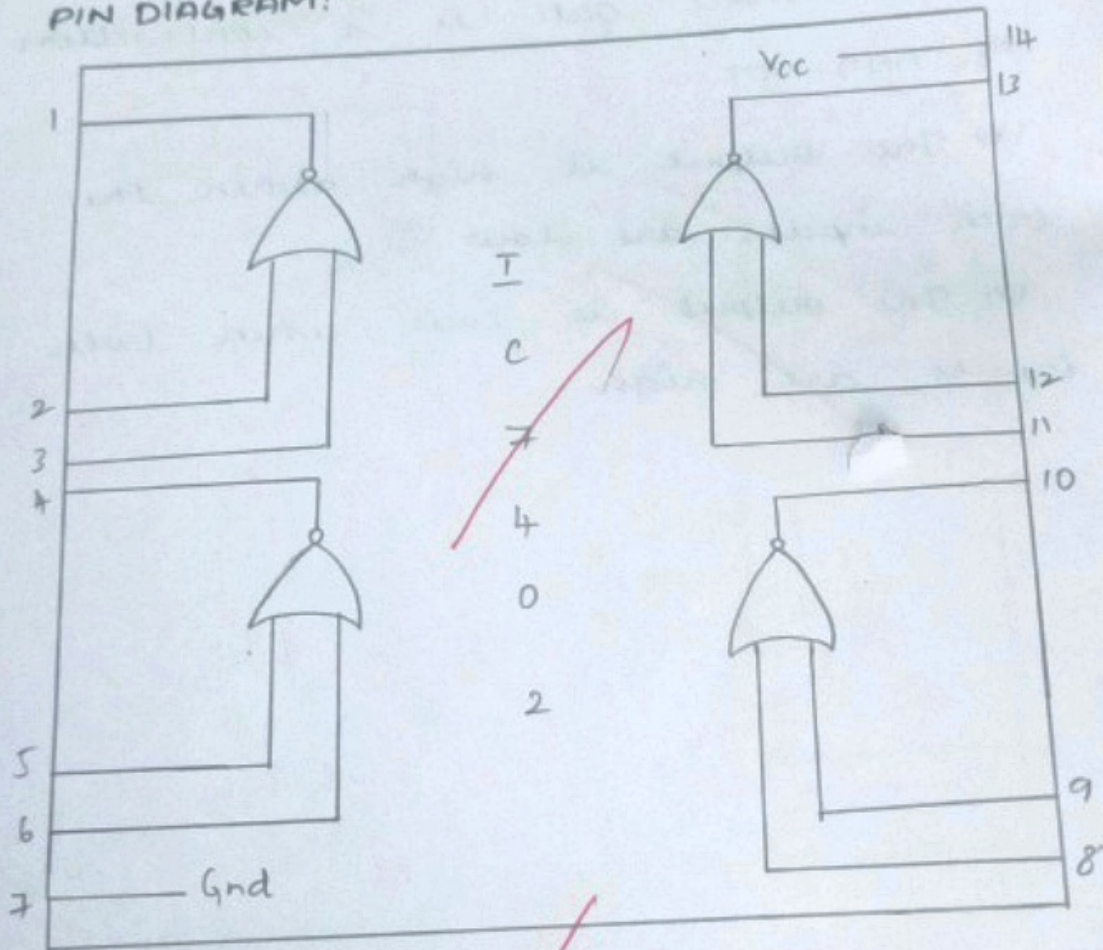


NOR GATE

SYMBOL:



PIN DIAGRAM:



TRUTH TABLE

A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate:

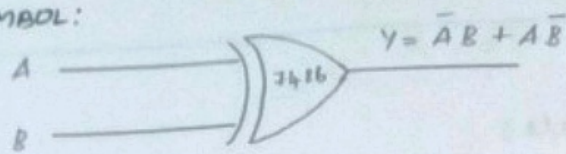
(*) The NOR gate is a contraction of OR-NOT.

(*) The output is high when both inputs are low.

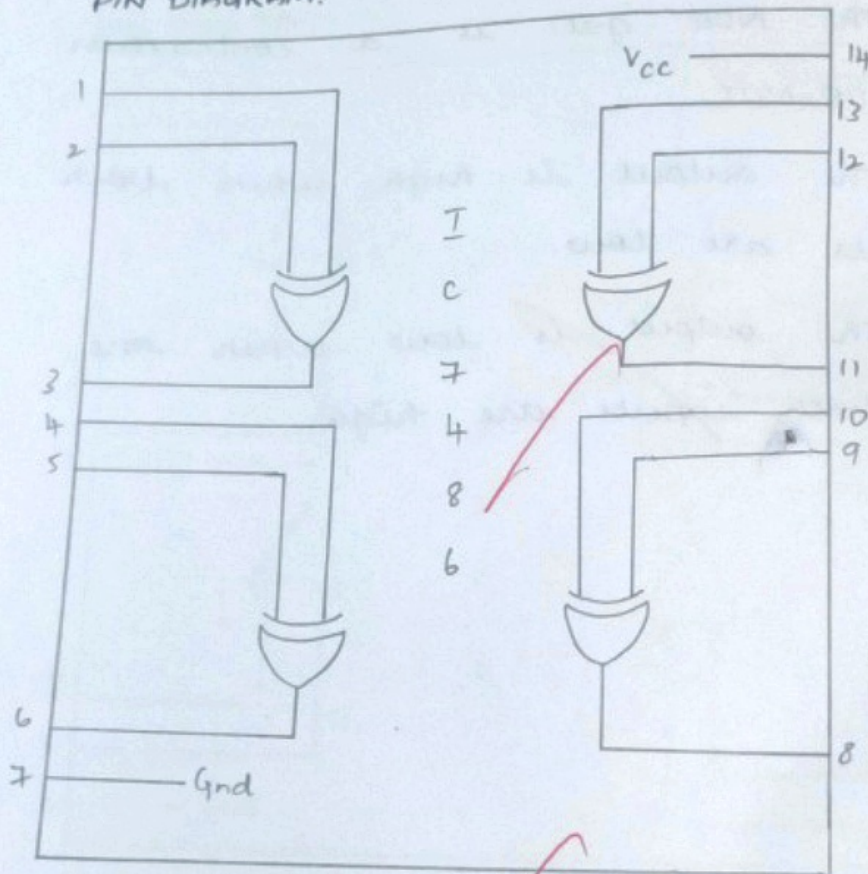
(*) The output is low when one or both inputs are high.

X-OR GATE

SYMBOL:



PIN DIAGRAM:



TRUTH TABLE:

A	B	$Y = \bar{A}B + A\bar{B}$
0	0	0
0	1	1
1	0	1
1	1	0

EX-OR gate:

(*) The output is high when any one of the input is high.

(*) The output is low when both the inputs are low and both the inputs are high.

①

RESULT:

Thus the logic gate was studied and the truth table was verified.

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EXP NO: &	VERIFICATION OF DEMORGAN'S THEOREM USING LOGIC GATES	DATE: 30/09/22
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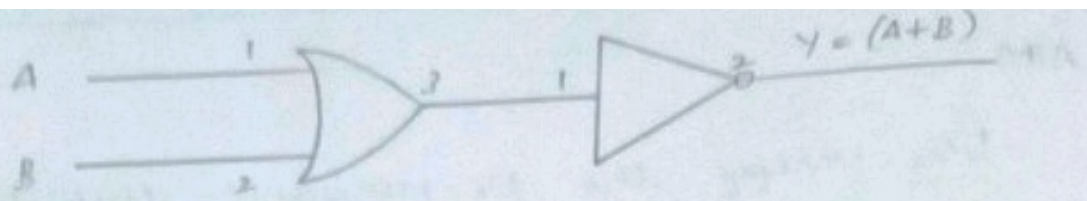
AIM:

To verify the De-Morgan's theorem using basic logic gates.

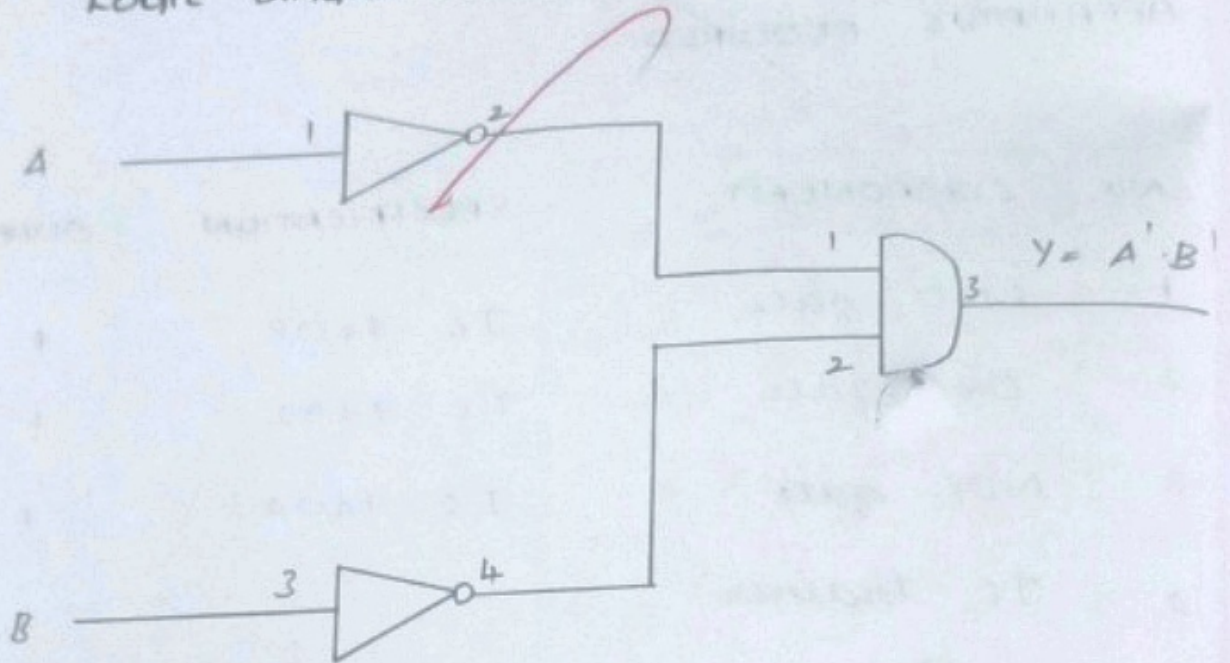
APPARATUS REQUIRED:

SNO	COMPONENT	SPECIFICATION	QUANTITY
1	AND gate	IC 7408	1
2	OR gate	IC 7432	1
3	NOT gate	IC 7404	1
4	IC Trainer kit	-	1
5	Connecting wires	-	14





LOGIC DIAGRAM: $(A' \cdot B')$



TRUTH TABLE:

A	B	$(A+B)'$	$A' \cdot B'$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

THEORY:

*) De-Morgan's suggested two theorems that form an important part of boolean algebra.

*) In the equation form, they are

$$(i) (A+B)' = A' \cdot B'$$

$$(ii) (A \cdot B)' = A' + B'$$

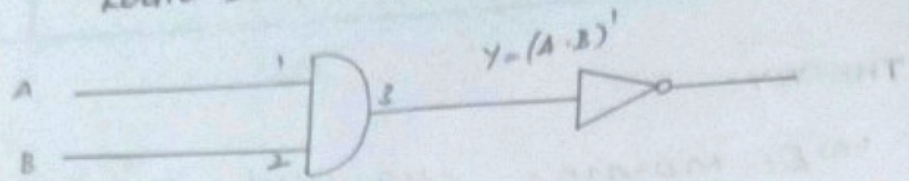
THEOREM 1:

$$(A+B)' = A' \cdot B'$$

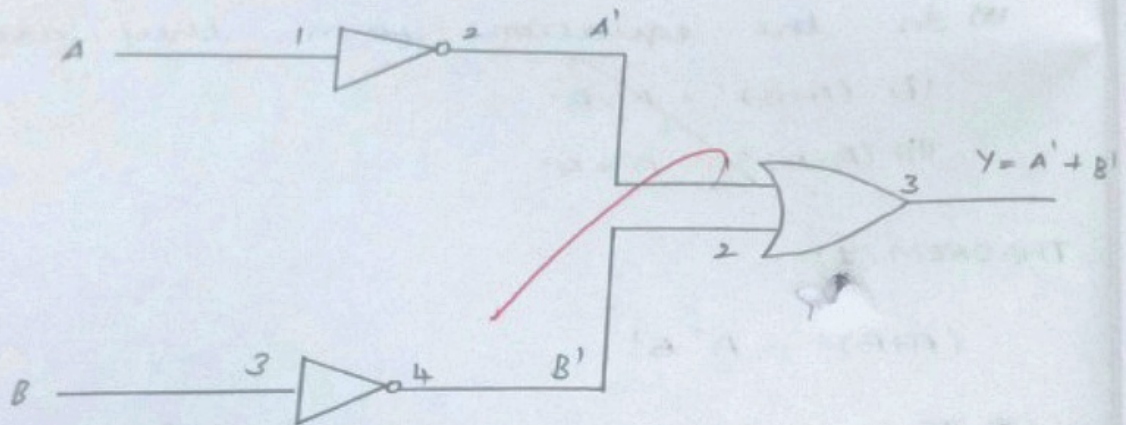
*) The complement of a sum is equal to the product of the complement.

*) The truth table shows that for all combinations of A and B, the two circuits give identical results.

LOGIC DIAGRAM: $(A \cdot B)'$



LOGIC DIAGRAM: $(A' + B')$



TRUTH TABLE:

A	B	$(A \cdot B)'$	$(A' + B')$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

THEOREM 2:

$$(A \cdot B)' = A' + B'$$

(*) The complement of the product is equal to the sum of the complement.

(*) The truth table 2 shows that for all combinations of A and B, the two circuit gives identical results.

PROCEDURE:

(*) Connection are given as per the circuit diagram.

(*) Logical input are given as per the circuit diagram.

(*) Observe the output and verify the truth table.

RESULT:

Thus the De-Morgan's theorem is implemented using logic gates and the truth table is verified.

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EXP NO: 3

DESIGN AND IMPLEMENTATION OF
ADDERS AND SUBTRACTORS

DATE:

30/09/22

AIM:

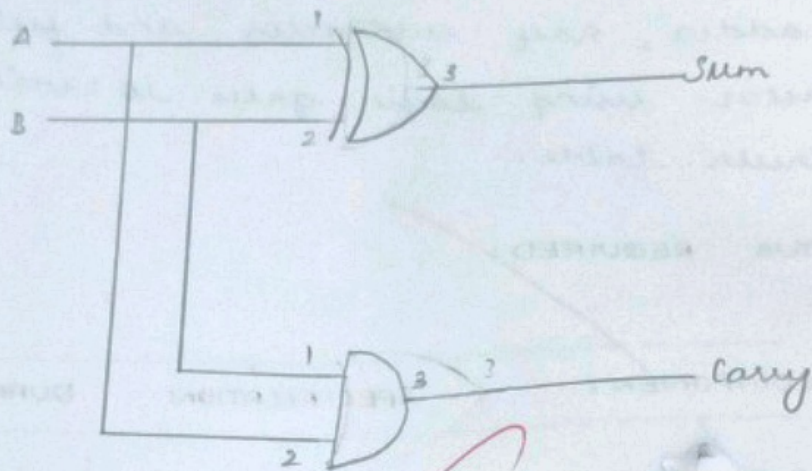
To design and construct half adder, full adder, half subtractor and full subtractor using logic gates to verify the truth table.

APPARATUS REQUIRED:

S.NO	COMPONENT	SPECIFICATION	QUANTITY
1	AND gate	IC 7408	1
2	OR gate	IC 7486	1
3	NOT gate	IC 7404	1
4	X-OR gate	IC 7432	1
5	Digital trainer kit	-	1
6	Path card	-	As required



HALF ADDER:



TRUTH TABLE

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

THEORY:

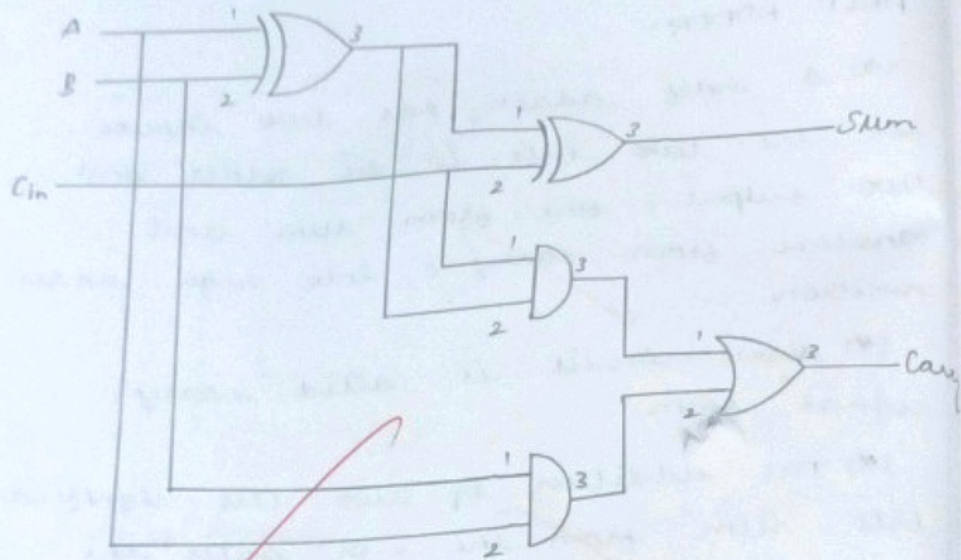
HALF ADDER:

(*) A half adder has two inputs for the two bits to be added and two outputs, one from sum and another from carry c into high order position.

(*) Above circuit is called carry signal form.

(*) The addition of two less significant bits sum from the X-OR gates the carry out from the AND gate.

FULL ADDER:



TRUTH TABLE:

A	B	C	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

FULL ADDER:

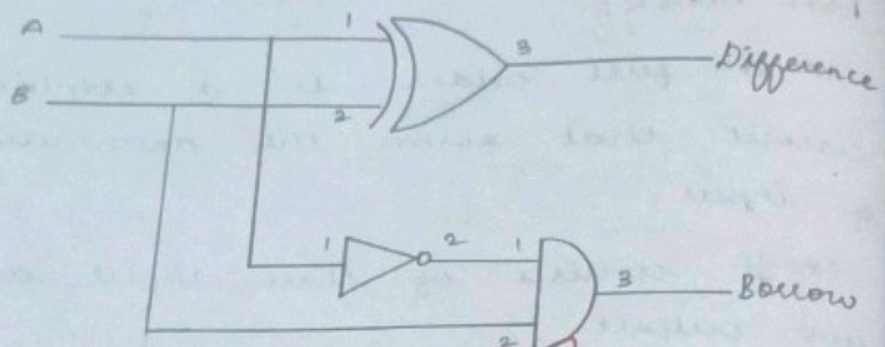
(*) A full adder is a combinational circuit that forms the arithmetic sum of input.

(*) It consists of three input and two output.

(*) A full adder is useful to add three bits at a time but a half adder cannot do so.

(*) In full adder sum output will be taken from X-OR gate carry output will be taken from OR gate.

HALF SUBTRACTOR:



TRUTH TABLE:

A	B	DIFFERENCE	BORROW
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

A red arrow points from the 'Borrow' column of the truth table towards the circuit diagram above.

HALF SUBTRACTOR:

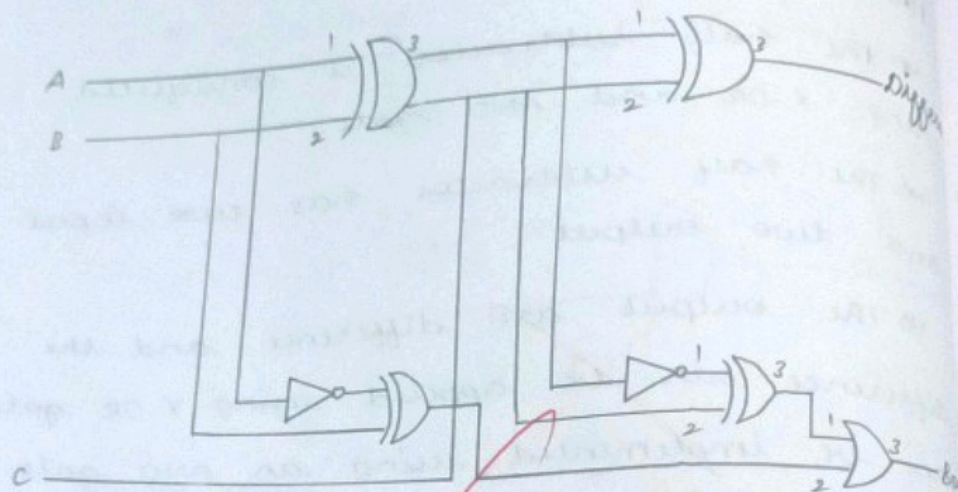
✱ The half subtractor is constructed using X-OR and AND gate.

✱ The half subtractor has two input and two output

✱ The output are difference and the difference can be applied using X-OR gate can be implemented using an AND gate and can inverter.

A		B		Difference	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	1	0	1	0	1
1	1	1	1	1	0

FULL SUBTRACTOR:



TRUTH TABLE:

A	B	C	DIFFERENCE	BORROW
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

FULL SUBTRACTOR:

(*) The full subtractor is combination of X-OR, AND, OR, NOT gates.

(*) In a full subtractor the logic circuit should have three input and two output.

(*) The two half subtractor put together gives a full subtractor.

(*) The half subtractor will be 1 and AB the output will be difference output of the half subtractor and the second term is the inverted difference output of first X-OR.



PROCEDURE:

- (*) Connections are given as per the circuit diagram
- (*) Logical inputs are given as per the circuit diagram
- (*) Observe the output and verify the truth table.

RESULT:

Thus the adder and subtractor circuit are implemented and verified using universal gates.

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